

# **Redes Neurais e Aprendizado Profundo**

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# **Multi-Query Attention and Grouped-Query Attention**

# Introduction

## *Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

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- Language models are expensive for inference primarily due to the memory bandwidth overhead from loading keys and values
- Multi-query attention (MQA) and Grouped-Query attention (GQA) reduce this overhead
  - Both methods provide a compromise between model capacity/quality and speed-up
- Ainslie et al.<sup>1</sup> propose to convert multi-head attention (MHA – the original Transformer architecture) models to multi-query and grouped-query models

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<sup>1</sup>Ainslie et al. *GQA Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. EMNLP, 2023

# Preliminaries

## *Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

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- MQA and GQA employ the Uptraining recipe
- Uptraining involves initializing a model from a pre-trained checkpoint
- Given the checkpoint, the recipe pre-trains for a further  $\alpha$  proportion of original pre-training steps
  - Uptraining by Ainslie et al.<sup>1</sup> employs the original pre-training setup and dataset

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<sup>1</sup>Ainslie et al. *GQA Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. EMNLP, 2023

# Multi-Query Attention

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- Multi-query attention (MQA) employ a single key-value head
- Going from Multi-head attention (MHA) to MQA reduces  $H$  key and value heads to a single key and value head
  - It reduces the size of the key-value cache and therefore amount of data that needs to be loaded by a factor of  $H$
  - It drastically speeds up decoder inference
- Unfortunately, MQA can lead to quality degradation and training instability<sup>1</sup>

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<sup>1</sup>Ainslie et al. *GQA Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. EMNLP, 2023



# Multi-Query Attention

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- Conversion from a multi-head model to a multi-query model takes place in two steps
  - Converting a checkpoint (i.e., a pre-trained MHA model)
  - Additional pre-training to allow the model to adapt to its new structure
- The projection matrices for key and value heads are mean-pooled into single projection matrices

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<sup>1</sup>Ainslie et al. *GQA Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. EMNLP, 2023

# Grouped-Query Attention

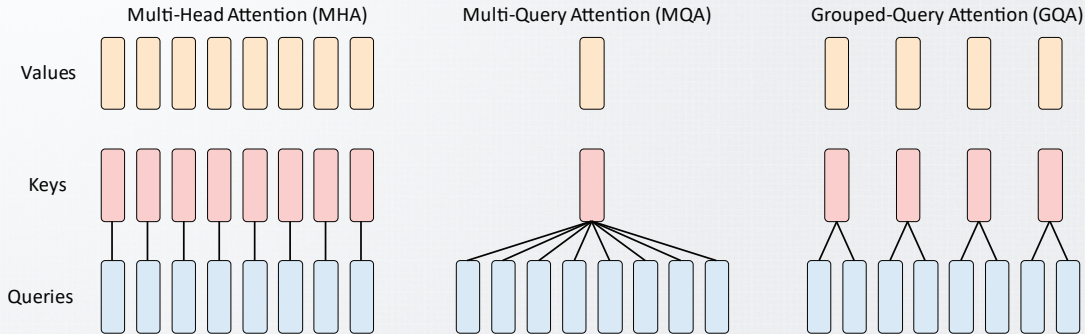
*Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

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- Grouped-query attention divides the query components into  $G$  groups
  - Each group  $g \in G$  shares a **single** key head and value head
- The GQA constructs each group key and value head by mean-pooling all the original heads within that group

# Overall Architectures

*Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

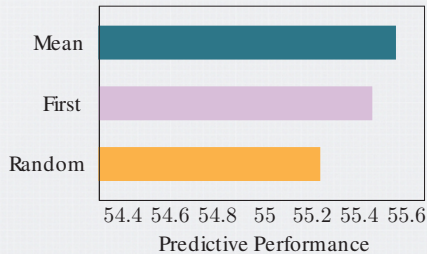
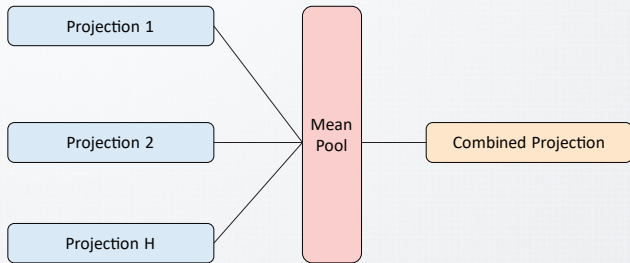




# Overall Architectures

## *Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

- Ainslie et al.<sup>1</sup> find this strategy works better than selecting a single key and value head or randomly initializing new key and value heads from scratch



<sup>1</sup>Ainslie et al. *GQA Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. EMNLP, 2023

# Overall Architectures

*Multi-query attention (MQA) & Grouped-Query Attention (GQA)*

